Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/01/2010
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Project Title:	Mechanisms Of High LET Radiation Induced Genomic Instability In The CNS		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 2/28/2010 per A. Chu/ARC (3/2009) NOTE: Start date changed for FY2007 to be 3/1/2007 as PI's anniversary date (per A. Chu 12/06)		
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Task Description:	The space radiation environment poses a range of potential health risks to astronauts. Exposure to high-energy protons and heavy ions within galactic cosmic rays or solar particle events can damage critical cellular targets leading to a number of deleterious effects. Space travel has also been found to elicit changes in the redox state of cells that may exacerbate these adverse effects. These risks may be particularly problematic to stem/precursor cells within the CNS, where exposure to irradiation may promote genomic instability which could result in cancer and/or cognitive dysfunction. In vitro and in vivo we have shown neural precursor cells are exquisitely sensitive to irradiation, undergoing a dose-dependent induction of apoptosis that is associated with a chronic oxidative stress. These effects are strongly correlated with a persistent inhibition of neurogenesis that may underlie cognitive deficits observed after cranial irradiation. We have also shown that human brain tumors contain a significant number of cells that express many of the same immature lineage-specific markers common to neural precursor cells, supporting a possible role for these cells in the development of brain tumors. Based on these observations, we hypothesize that neural precursor cells exposed to high energy ions will be particularly susceptible to genomic instability and in a manner that depends upon redox state. Genomic instability is a process believed to drive cancer progression, and recently, redox state has been implicated in the induction and perpetuation of genomic instability. Redox state has been found to be a critical determinant in regulating the response of the CNS to a variety of damaging treatments, including irradiation. Changes in the redox environment are often associated with alterations in cellular metabolism which can promote oxidative stress, and these types of changes are common to tumors including those within the CNS. Because exposure to radiation has been found to cause genomic instability, and since oxidative stress is associated with the radiation response of neural precursor cells, it follows that changes in redox state might constitute a biochemical mechanism for modifying the radioresponse of the CNS. We are proposing a series of studies to test the overall hypothesis that alterations in cellular redox state promote genomic instability that can adversely impact the CNS upon exposure to high LET radiation. While the development of genomic instability may predispose neural precursor cells to transformation events in the CNS, the unstable phenotype may also lead to radioresistance that changes the phenotypic fate of neural precursor cells that impairs neurogenesis. We propose that changes in redox state play a contributory if not causative role in regulating the radiation response of neural stem cells in vitro and in vivo; changes that will foster the onset of radiation-induced genomic instability, radioresistance and altered neurogenesis. Using an in vitro and in vivo approach the goals of our project will be addressed in the following 3 specific aims: Specific aim 1: Elucidate the dose-response relationship for the induction of genomic instability in neural precursor cells cultured in vitro or isolated from mouse brain after exposure to protons and iron ions. Specific aim 2: Determine if chronic changes in cellular redox state alter the dose response relationship determined above and predispose neural precursor cells to proton- and iron ion-induced genomic instability. Specific aim 3: Determine if differentiation of genomically unstable neural precursor cells is altered in vitro, or if neurogenesis is altered in vivo when unstable precursor cells are implanted into the brains of control or irradiated mice. Task Objective: To determine whether changes in redox state adversely impact the CNS. We will assess whether oxidative stress alters the radiosensitivity of neural precursors and predisposes them radiation-induced genomic instability. Experiments will ascertain whether such changes alter precursor cell plasticity in a manner that compromises their normal function. These studies will promote an understanding of the space radiation effects on the CNS. Data collected from these studies will improve both our understanding of the basic biology underlying CNS radiation effects, and in assessing risks to astronauts engaged in space travel. This work will establish the extent to which redox related changes derived from environmental stressors (oxidative injury) and/or damaging agents (ionizing radiation) are problematic to astronauts and whether specific countermeasures can be developed to minimize such risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will shed light on many of the basic biological processes that are associated with exposure of the CNS to ionizing radiation. As such, our work is directly relevant to those involved in the treatment of CNS tumors via radiotherapy. Such individuals include clinicians, basic scientists and even patients, all of whom have a vested interest in minimizing side-effects associated with the exposure of normal tissue to ionizing radiation. Outside of the clinical relevance, our work should foster a general interest to those members of the public supporting the long-term goals of NASA involving the expoloration and colonization of space.
Task Progress:	We are now at the end of the funding period for this grant and are pleased with much of our progress. Over this funding period we have undertaken a range of in vitro studies that have uncovered a number of radioresponsive endpoints in neural stem and precursor cells that are likely to impact genomic instability. Many of our major accomplishments are summarized below: 1) In vitro studies using hippocampal precursors have revealed that endogenous oxidative stress in the form of hydrogen peroxide increases the sensitivity of these cells to X-rays. 2) In vitro studies using hippocampal precursors have revealed that exogenous oxidative stress increases the sensitivity of these cells to X-rays and to 250 MeV protons. 3) In vitro studies using hippocampal precursors have revealed that endogenous oxidative stress in the form of superoxide anion actually decreases the sensitivity of these cells to X-rays. These findings illuminate an important point regarding the dual nature of oxidative stress, namely, it can be good or bad depending on type of reactive species involved and the microenvironmental context in which the stress occurs. Thus, while hydrogen peroxide sensitizes neural precursor cells to irradiation, superoxide protects them. 4) In vitro studies have shown that perturbation of mitochondrial electron transport increases oxidative stress and enhances the sensitivity of neural precursors to irradiation. 5) Comparisons between neural precursor cells and brain tumor cells have shown that these cells share similar redox properties after irradiation. Neural precursor cells and gliomas show marked increases in oxidative stress after irradiation, an effect not found when normal brain cells were analyzed after irradiation. 6) Neural precursor cells and brain tumor cells express a number of shared phenotypic markers. The shared expression of immature markers suggests that a fraction of brain tumors contain cells similar to neural precursor cells.

	7) Normal neural stem and precursor cells exhibit CD133 expression, a marker reported to be specific for brain tumor stem cells. 8) Dose fractionation of normal subventricular stem and precursor cells using either x-rays or heavy ions selects for a radioresistant subpopulation enriched for CD133 positive cells. 9) Cell survival and neurosphere assays reveals neural stem cells to be more radioresistant than their precursor cell progeny.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Fike JR, Rosi S, Limoli CL. "Neural precursor cells and central nervous system radiation sensitivity." Semin Radiat Oncol. 2009 Apr;19(2):122-32. PMID: 19249650 , Apr-2009
Books/Book Chapters	Globus RK, Caiozzo VJ, Acharya M, Fike JR, Limoli CL. "Redox regulation of stem cell compartments: The convergence of radiation-induced normal tissue damage and oxidative stress." in "Oxidative Stress in Cancer Biology and Therapy." Ed. D.R. Spitz, K.J. Dornfeld, K. Krishnan, D. Gius. New York : Springer, 2012. p.169-192., Jan-2012